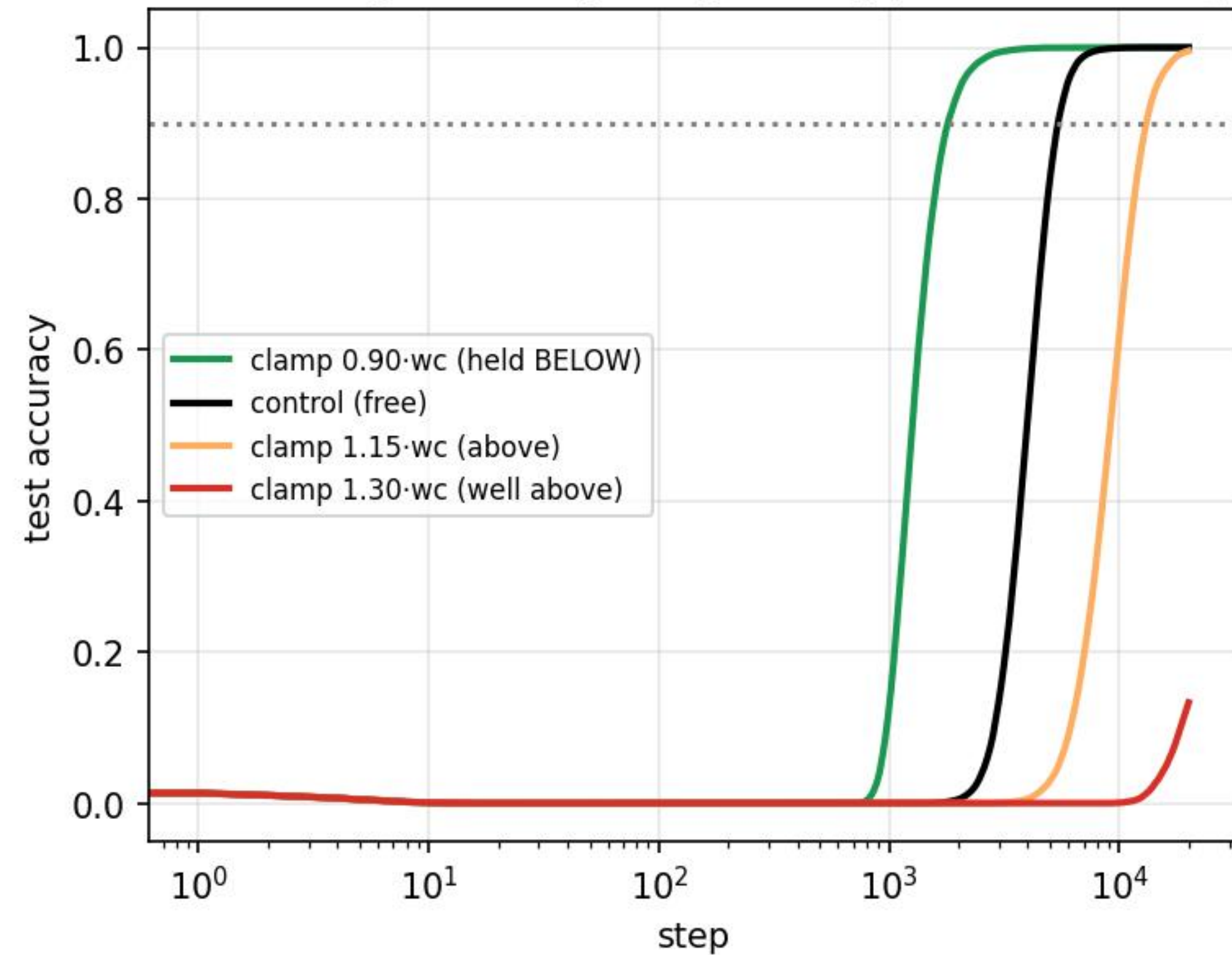
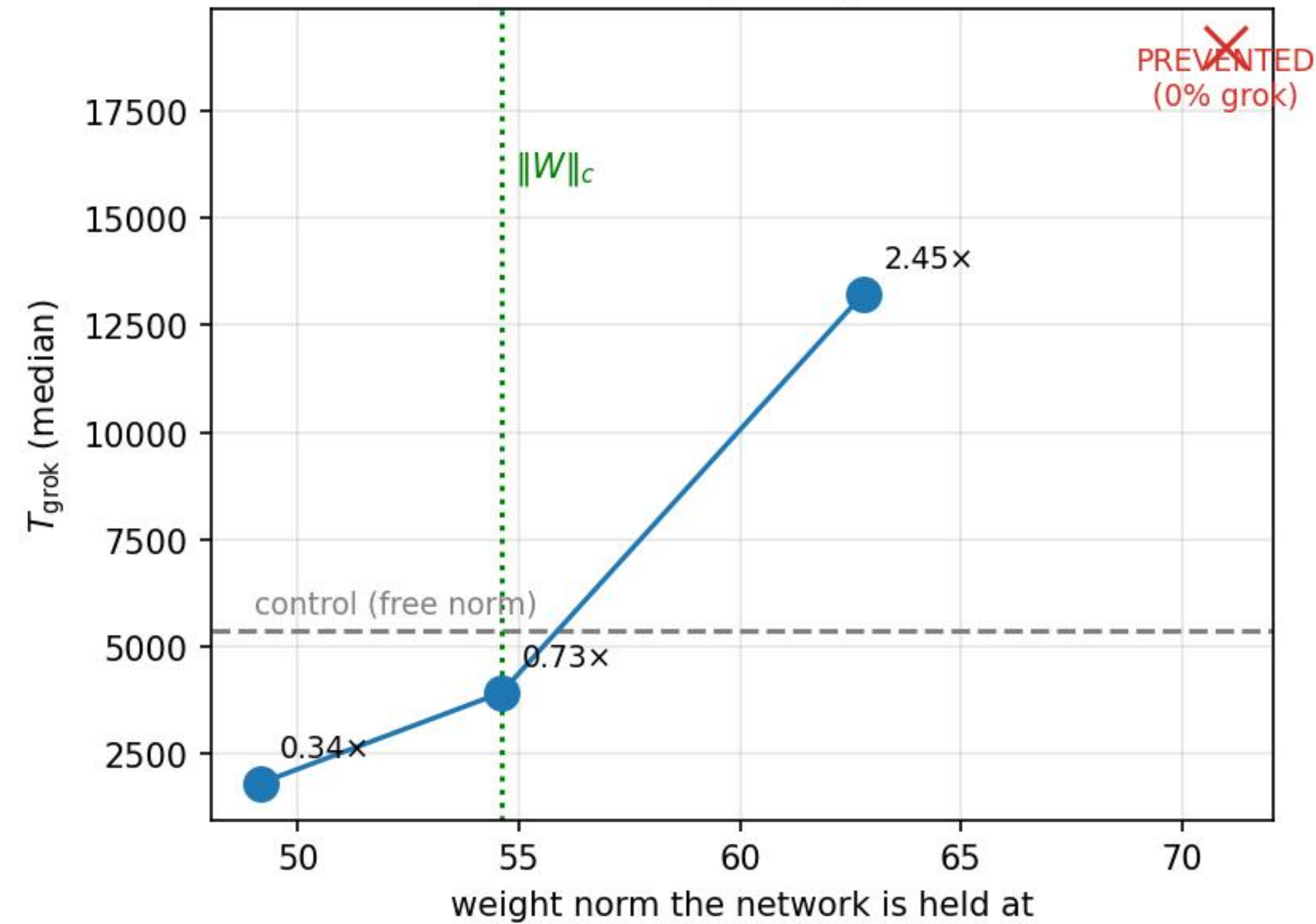


A. The weight norm gates grokking (both directions)



B. Causal dose-response: lower norm $\rightarrow$ faster grok,
higher $\rightarrow$ delayed $\rightarrow$ prevented



C. Sustained $\gg$ transient
(a single kick washes out; holding it matters)

